

Cache Hit/Miss

1. Does the memory access with PC 0x401dc9 and address 0x47ea85d37f result in a cache hit or cache miss for the lbm workload and Parrot replacement policy?
Ans: **Cache Miss**
Type: **Cache Hit/Miss**
2. For the memory access with PC 0x4037ba and address 0xa3a0df59c8, does the LRU replacement policy result in a cache hit or miss?
Ans: **Cache Hit**
Type: **Cache Hit/Miss**
3. When the LRU replacement policy accesses address PC 0x405832 and address 0x2ad51d2d9d on the astar workload, does the cache hit or miss?
Ans: **Cache Miss**
Type: **Cache Hit/Miss**
4. For the cache access with PC 0x401e31 and address 0x35e798a637f on the lbm workload with MLP replacement policy, does the cache hit or miss?
Ans: **Cache Miss**
Type: **Cache Hit/Miss**
5. Does the memory access with PC 0x4090c3 and address 0x1996629d95c result in a cache hit or cache miss for the astar workload and lru replacement policy?
Ans: **Cache Miss**
Type: **Cache Hit/Miss**
6. For the memory access with PC 0x4090c3 and address 0x3a4c39acd9d, does the parrot replacement policy result in a cache hit or miss for workload astar?
Ans: **Cache Hit**
Type: **Cache Hit/Miss**
7. When the LRU replacement policy accesses address PC 0x409270 and address 0x2bfd401c63f on the astar workload, does the cache hit or miss?
Ans: **Cache Miss**
Type: **Cache Hit/Miss**
8. For the cache access with PC 0x405832 and address 0x210528d2d9d on the astar workload with beladys replacement policy, does the cache hit or miss?
Ans: **Cache Miss**
Type: **Cache Hit/Miss**
9. Does the memory access with PC 0x401804 and address 0x1edc2b90cc7 result in a cache hit or cache miss for the lbm workload and belady's replacement policy?
Ans: **Cache Hit**
Type: **Cache Hit/Miss**
10. For the memory access with PC 0x401804 and address 0x3b30bc0c1ca, does the parrot replacement policy result in a cache hit or miss for workload lbm?
Ans: **Cache Miss**
Type: **Cache Hit/Miss**
11. Does the memory access with PC 0x401e31 and address 0x17068839426 result in a cache hit or cache miss for the lbm workload and lru replacement policy?

Ans: **Cache Hit**

Type: **Cache Hit/Miss**

12. Does the memory access with PC 0x401ea1 and address 0x33e58171806 result in a cache hit or cache miss for the lbm workload and mlp replacement policy?

Ans: **Cache Miss**

Type: **Cache Hit/Miss**

13. What happens when there is a memory access with PC 0x4037c1 and address 0xa3a0df544d for the mcf workload and mlp replacement policy? Does this result in a cache hit or cache miss

Ans: **Cache Miss**

Type: **Cache Hit/Miss**

14. Does the memory access with PC 0x4037aa and address 0x1b73be82e3f result in a cache hit or cache miss for the mcf workload and parrot replacement policy?

Ans: **Cache Miss**

Type: **Cache Hit/Miss**

15. With the cache access for PC 0x4037ba and address 0xa3a0df8f14 on mcf workload using LRU replacement policy, will the final decision be a cache hit or cache miss?

Ans: **Cache Miss**

Type: **Cache Hit/Miss**

16. Does the memory access with PC 0x401cfb and address 0x2ae259bc1f3 result in a cache hit or cache miss for the lbm workload and Parrot replacement policy?

Ans: **Cache Miss**

Type: **Cache Hit/Miss**

17. Does the memory access with PC 0x40187e and address 0x3cef5f5eb19 result in a cache hit or cache miss for the lbm workload and Parrot replacement policy?

Ans: **Cache Miss**

Type: **Cache Hit/Miss**

18. Does the memory access with PC 0x401e47 and address 0xbc9e55c5b0 result in a cache hit or cache miss for the lbm workload and Parrot replacement policy?

Ans: **Cache Miss**

Type: **Cache Hit/Miss**

19. Does the memory access with PC 0x401705 and address 0x3b30bc0c1c8 result in a cache hit or cache miss for the lbm workload and Parrot replacement policy?

Ans: **Cache Miss**

Type: **Cache Hit/Miss**

20. For the memory access with PC 0x4037ba and address 0xa3a0df59c8, does the LRU replacement policy result in a cache hit or miss?

Ans: **Cache Hit**. MCF workload - 232 hits, 64 misses.

Type: **Cache Hit/Miss**

21. When the LRU replacement policy accesses address PC 0x405832 and address 0x2f6b92aa544 on the astar workload, does the cache hit or miss?

Ans: **Cache Miss**. 1 hit, 34 misses

Type: **Cache Hit/Miss**

22. Under the Belady policy, does PC 0x4090c3 and address 0x1996629d95c result in a hit or miss in the astar workload?
Ans: **Cache Hit**.
Type: **Cache Hit/Miss**
23. In the lbm workload under the LRU replacement policy, does accessing address 0x2fdf757709d at PC 0x401e89 lead to a cache hit or miss?
Ans: **Cache Miss**.
Type: **Cache Hit/Miss**
24. For PC 0x4037ba and address 0x29f422b43a0 in the mcf workload, does PARROT policy result in a cache hit or cache miss?
Ans: **Cache Hit**.
Type: **Cache Hit/Miss**
25. In the lbm workload with the LRU policy, when PC=0x401e31 accesses memory address=0x17068839426, is it a cache hit or miss?
Ans: **Cache Hit**
Type: **Cache Hit/Miss**
26. Does the memory access with PC 0x401e31 and address 0x35e798a637f result in a cache hit or cache miss for the lbm workload and Parrot replacement policy?
Ans: **Cache Miss**.
Type: **Cache Hit/Miss**
27. For the memory access with PC 0x405832 and address 0x2ad51d2d9d, does the LRU replacement policy result in a cache hit or miss?
Ans: **Cache Miss**.
Type: **Cache Hit/Miss**
28. When the LRU replacement policy accesses PC 0x405832 and address 0x2ad51d2d9d on the astar workload, does the cache hit or miss?
Ans: **Cache Miss**.
Type: **Cache Hit/Miss**
29. For the cache access with PC 0x401e31 and address 0x35e798a637f on the lbm workload with MLP replacement policy, does the cache hit or miss?
Ans: **Cache Miss**.
Type: **Cache Hit/Miss**
30. Does the memory access with PC 0x4037aa and address 0x3cab60f39f3 result in a cache hit or cache miss for the mcf workload and belady's replacement policy?
Ans: **Cache Hit**
Type: **Cache Hit/Miss**

Miss Rate

31. What is the miss rate for PC 0x4037ba on the mcf workload with PARROT replacement policy?
Ans: **Miss rate for PC=0x4037ba: 0.03504614926558158**
Type: **Miss Rate**
32. How does the LRU replacement policy perform in terms of miss rate and cache hit rate on the astar workload? (what does this even mean?)

Ans: **Cache Miss Rate = 94.91%; Cache Hit Rate = 5.09%.**

Type: **Miss Rate**

33. What is the miss rate for the mcf workload with PARROT replacement policy?

Ans: **58.81% miss rate.**

Type: **Miss Rate**

34. In lbm + BELADY, between PC=0x401ed1 and PC=0x401e31, which has a higher miss rate?

Ans: **PC=0x401e31 has a higher miss rate.** PC=0x401ed1 miss rate: 0.4796653497559842; PC=0x401e31 miss rate: 0.9892498247254031.

Type: **Miss Rate**

35. For the lbm workload, focusing on PC=0x401736, which policy LRU or PARROT has a lower miss rate?

Ans: **LRU.** miss rate: 0.9874476987447699;

Type: **Miss Rate**

36. In the astar workload with Belady, what is the miss rate for PC=0x405832?

Ans: **0.5650817298907925**

Type: **Miss Rate**

37. What is the miss rate for PC 0x401dc9 on the lbm workload with the LRU replacement policy?

Ans: **Miss rate for program_counter 0x401dc9: 98.79%**

Type: **Miss Rate**

38. What is the miss rate for PC 0x4037aa on the mcf workload with PARROT replacement policy?

Ans: **Miss rate for program_counter 0x4037aa: 99.43%.**

Type: **Miss Rate**

39. What is the miss rate for PC 0x405832 on the astar workload with Belady replacement policy?

Ans: **Miss rate for program_counter 0x405832: 56.51%**

Type: **Miss Rate**

40. What is the average miss rate for the mcf workload using the LRU policy?

Ans: **72.75% miss rate.**

Type: **Miss Rate**

Policy Comparison

41. Across the entire lbm workload, which replacement policy Belady, LRU, PARROT, or MLP shows the highest number of misses?

Ans: **MLP or Parrot.** Number of Misses: Belady = 18743, LRU = 44423, MLP = 44475, Parrot = 44475

Type: **Policy Comparison**

42. Which replacement policy performs best on the astar workload?

Ans: **Best policy for astar is belady,** with miss rate = 0.608255628837844

Type: **Policy Comparison**

43. Compare the miss rates for PC 0x4037c1 under all four replacement policies, i.e, LRU, MLP, Parrot and Belady's on the mcf workload. Which one performs the worst?

Ans: **MLP**. LRU = 28.02%; MLP = 48.37%; Parrot = 15.25%; Belady = 2.25%

Type: **Policy Comparison**

44. Compare the miss rates for PC 0x40182a under LRU and Parrot replacement policies on the lbm workload. Which one performs better?

Ans: **LRU**. LRU = 98.91%; Parrot = 98.95%

Type: **Policy Comparison**

45. Which replacement policy performs best on the lbm workload and which is the worst?

Ans: **Best = Belady, Worst = All the rest.**

Type: **Policy Comparison**

46. Which replacement policy has the highest miss rate for PC 0x409270 on the astar workload?

Ans: **MLP**.

Type: **Policy Comparison**

47. What is the ranking of the four replacement policies (LRU, Parrot, MLP, Belady) based on the cache miss rate from highest to lowest on the mcf workload?

Ans: **LRU > MLP > Parrot > Belady.**

Type: **Policy Comparison**

48. What is the ranking of the four replacement policies (LRU, Parrot, MLP, Belady) based on the cache hit rate from highest to lowest on the mcf workload?

Ans: **Belady > Parrot > MLP > LRU.**

Type: **Policy Comparison**

49. Which replacement policy (LRU, MLP, Parrot or Belady) has the lowest miss rate for PC 0x409270, on the astar workload?

Ans: **Belady**. LRU = 98.10%; MLP = 98.26%; Parrot = 96.15%; Belady = 95.98%.

Type: **Policy Comparison**

50. Which replacement policy has the lowest miss rate for PC 0x409270 on the lbm workload?

Ans: **Best policy for PC=0x409270 in astar is belady**, with miss rate = 0.9597988077496274

Type: **Policy Comparison**

51. Which replacement policy in the astar workload has the highest overall miss rate?

Ans: **lru**, with miss rate=0.9490988173754833

Type: **Policy Comparison**

52. Which workload has the highest cache miss rate using the MLP policy?

Ans: **LBM**.

Type: **Policy Comparison**

53. Which replacement policy yields the lowest overall miss rate on the lbm workload?

Ans: **Belady** with miss rate= 0.7358316566063045

Type: **Policy Comparison**

54. Which replacement policy performs worse on the mcf workload, Parrot or MLP?

Ans: **MLP**. MLP miss rate = 82.82%, Parrot miss rate = 58.81%

Type: **Policy Comparison**

55. Which replacement policy has the highest miss rate for PC 0x4037c1, on the mcf workload?

Ans: **MLP**. LRU miss rate = 28.02%; MLP miss rate = 48.37%; Parrot miss rate = 15.25%; Belady miss rate = 2.25%.

Type: **Policy Comparison**

Count

56. How many times does PC 0x401e47 occur in the lbm workload under the PARROT policy?

Ans: **8329**

Type: **Count**

57. How many times did PC 0x405832 appear in the trace for the astar workload with the LRU policy?

Ans: **Number of times program_counter 0x405832 occurred in the trace: 71149**

Type: **Count**

58. How many times does the memory access with PC 0x4037ba and address 0xa3a0df3d9d result in a cache miss on the mcf workload using the MLP replacement policy?

Ans: **Number of cache misses for PC 0x4037ba and memory address 0xa3a0df3d9d: 115**

Type: **Count**

59. How many times does the memory access with PC 0x405828 and address 0x35520d06f6 result in a cache hit on the astar workload using the Belady replacement policy?

Ans: **Number of cache hits for PC 0x405828 and memory address 0x35520d06f6: 3.**

Type: **Count**

60. In the mcf workload, does PC=0x4037ba access the same address=0xa3a0df8d44 multiple times? How many hits vs. misses?

Ans: **Policy=lru, repeated accesses=565, hits=514, misses=51; Policy=belady, repeated accesses=565, hits=563, misses=2; Policy=mlp, repeated accesses=505, hits=341, misses=164; Policy=parrot, repeated accesses=505, hits=505, misses=0**

Type: **Count**

Arithmetic

61. At PC 0x401cfb in the lbm workload, which policy has the lowest average reuse distance of inserted addresses, indicating suboptimal eviction choices?

Ans: **MLP or Parrot**. LRU = 3183.8721149317003; MLP = 3182.9061893203884; Parrot = 3182.9061893203884; Belady = 3183.837376354216.

Type: **Arithmetic**

62. What is the average evicted reuse of PC 0x40170a for the lbm workload with MLP replacement policy?

Ans: **Average evicted reuse distance for PC 0x40170a: 3595.6429149797573.**

Type: **Arithmetic**

63. Under lbm + PARROT, which PC has the longest average reuse distance?

Ans: **PC= 0x400d81, avg reuse distance= 4256.0**

Type: **Arithmetic**

64. Under astar + MLP, which memory address is evicted most frequently?

Ans: **Address= 0x18d77fe567e**, eviction count= 66

Type: **Arithmetic**

65. For the mcf workload with PARROT, which PC repeatedly hits the same address consecutively the most times?

Ans: **PC=0x4037ba**, address=0xa3a0df8d44 with 505 consecutive hits.

Type: **Arithmetic**

66. In the mcf workload using PARROT, which PC generates the most misses for address=0xa3a0df8d44?

Ans: **PC=0x4013a1, misses=14**

Type: **Arithmetic**

67. Which policy in the astar workload has the smallest average evicted reuse distance for PC=0x405832?

Ans: **LRU**. Average Access Reuse Distance for LRU = 10199.798226257572; MLP = 49651.60660809306; Parrot = 51214.96926432372; Belady = 45268.20821093761.

Type: **Arithmetic**

68. What is the average evicted reuse distance for the most frequently evicted memory address in mcf workload with Parrot replacement policy?

Ans: The average evicted reuse distance for memory address 0x3f7f4ea2d44, which is the most frequently evicted, is **110604.10394265233**.

Type: **Arithmetic**

69. How many times does the memory access with PC 0x401e89 and address 0x2fdf757709d result in a cache hit on the lbm workload using the LRU replacement policy?

Ans: **Cache access with PC 0x401e89 and memory address 0x2fdf757709d results in 0 cache hits and 1 cache miss.**

Type: **Arithmetic**

70. For astar + Parrot, which memory address is evicted most frequently?

Ans: 0x286af5ff319

Type: **Arithmetic**

Trick Questions

71. In the mcf workload with Belady, does PC=0x4095ab accessing address=0x1edc2b9109d trigger any eviction?

Ans: **No matching record found. TRICK QUESTION**

Type: **Trick Question**

72. Does the memory access with PC 0x4037aa and address 0x1b73be82e3f result in a cache hit or cache miss for the lbm workload and parrot replacement policy?

Ans: **No matching record found. TRICK QUESTION - mcf workload**

Type: **Trick Question**

73. Does the memory access with PC 0x4037aa and address 0x1b73be82e3d result in a cache hit or cache miss for the lbm workload and parrot replacement policy?

Ans: **No matching record found.** TRICK QUESTION - Address doesn't exist

Type: **Trick Question**

74. Why does the Parrot replacement policy perform better than LRU on the Ibm workload when comparing the cache miss rate?

Ans: **It does not.** TRICK QUESTION

Type: **Trick Question**

75. For astar workload with LRU policy where PC=0x409270 and memory address=0x2bfd401f35e, why does a cache miss occur but the "is_miss" value is 0?

Ans: **It does not.** TRICK QUESTION

Type: **Trick Question**

Microarchitecture Concepts

1. Suppose for some benchmark and replacement policy, the pair - PC "A" and memory address "X", occurs 50 times in the database. It results in a cache hit 40 times and a cache miss 10 times. Would you categorize the PC, memory access pair as a cache hit or a cache miss?
Type: **Microarchitecture**
2. Consider a 64-bit CPU with a 4-way set associative cache with cache block size of 64 bytes and 2 MB size. Given a memory address - 0x35520d06f6, what is the block offset value in decimal, index in decimal and the tag address in hexadecimal?
Type: **Microarchitecture**
3. Consider the following two scenarios - 1) replacement policy A is simulated over 100,000 cache accesses to LLC, which results in 35,000 cache hits, and, 2) replacement policy B is simulated over 50,000 cache accesses to L2 cache which results in 15,000 cache misses. Which of the two policies (A or B) performs better in terms of miss rate?
Type: **Microarchitecture**
4. How does the cache miss rate change by increasing cache size? How does it change if we increase the cache by the same amount in two different ways - 1) by increasing the number of ways, and, 2) by increasing the number of sets?
Type: **Microarchitecture**
5. Consider a piece of code that has a loop running 100,000 times. The system only has L1 cache with 64 byte block size. There are accesses to memory addresses "X", "X+64", "X+128" and "X+192" (decimal format) in one iteration of the loop and no other memory access. Which type of cache: direct-mapped or fully associative, would give better performance in terms of cache misses and in turn IPC?

Type: **Microarchitecture**

Code Generation

6. How many times does the memory access with PC 0x405828 and address 0x35520d06f6 result in a cache hit on the astar workload using the Belady replacement policy? Can you write the code to find out the same. The database is a dataframe for the eviction data of **the astar workload and Belady policy. Each dataframe has the following**

relevant columns - "program_counter" (PC, hex format), "memory_address" (hex format) and "is_miss" (1=miss, 0=hit).

Type: **Code**

7. What is the miss rate for PC 0x4037aa on the mcf workload with PARROT replacement policy? Can you write the code to find out the same. The database is a dataframe for the eviction data of the mcf workload and PARROT policy. Each dataframe has the following relevant columns - "program_counter" (PC, hex format), "memory_address" (hex format) and "is_miss" (1=miss, 0=hit).

Type: **Code**

8. Which replacement policy has the highest miss rate for PC 0x409270 on the astar workload? Can you write the code to find out the same. The database contains dataframes for the eviction data of the astar workload and belady, lru, mlp and parrot replacement policies, which are indexed as "loaded_data[astar_evictions_<policy>]", eg: loaded_data[astar_evictions_lru]. Each dataframe has the following relevant columns - "program_counter" (PC, hex format), "memory_address" (hex format) and "is_miss" (1=miss, 0=hit).

Type: **Code**

9. How many times does the memory access with PC 0x4037ba and address 0xa3a0df3d9d result in a cache miss on the mcf workload using the MLP replacement policy? Can you write the code to find out the same. The database is a dataframe for the eviction data of the mcf workload and MLP policy. Each data frame has the following relevant columns - "program_counter" (PC, hex format), "memory_address" (hex format) and "is_miss" (1=miss, 0=hit).

Type: **Code**

10. A dataframe contains the eviction data of the astar workload and Belady policy. It has the following relevant columns - "program_counter" (PC, hex format), "memory_address" (hex format) and "is_miss" (1=miss, 0=hit). Could you give the code to plot the number of hits and number of misses for PC 0x405832 and memory address 0x2ad51d2d9d?

Type: **Code**

Replacement Policy

11. What are the potential trade-offs between the LRU and Parrot policies in terms of miss rate?

Type: **Replacement Policy**

12. Why does the PARROT policy achieve a higher hit rate than LRU for PCs associated with recursive functions in astar?

Type: **Replacement Policy**

13. Why does the astar workload have a high miss rate across the LRU, MLP, and PARROT replacement policies?

Type: **Replacement Policy**

14. Analyze how eviction line reuse distance changes across different policies (LRU, MLP, PARROT) for PC 0x4037ba in the mcf workload. Explain why.

Type: **Replacement Policy**

15. Analyze the relationship between the accessed address reuse distance and the cache hit rate.

Type: **Replacement Policy**

Workload Analysis

16. What is the miss rate for PC 0x4037aa on the mcf workload with PARROT and Belady's replacement policy? Why is it so high?

Type: **Workload Analysis**

17. For the lbm workload, consider PC 0x401cfb. What is the average evicted reuse distance for belady's and LRU replacement policies for this PC? Analyze the nature of the replacement policies and justify why a certain policy has a longer eviction reuse distance than the other.

Type: **Workload Analysis**

18. Why does the PARROT policy achieve a higher hit rate than LRU for PCs associated with recursive functions in astar?

Type: **Workload Analysis**

19. For which workload did all of the policies struggle with the most? Why do you think this is?

Type: **Workload Analysis**

20. Analyze how eviction line reuse distance changes across different policies (MLP, LRU, PARROT) for PC 0x4037ba for the mcf workload.

Type: **Workload Analysis**

Semantic Analysis

21. For the lbm workload, PC 0x401dc9 results in a cache miss. Looking at the assembly code, what might be the reason for this?

Type: **Semantic Analysis**

22. The miss rate for PC 0x401dc9 on the lbm workload with the LRU replacement policy is 98.79%. Look at the assembly code to analyze what could be the reason for the same.

Type: **Semantic Analysis**

23. In the mcf workload, for any replacement policy, PC=0x4037ba accesses the same address=0xa3a0df8d44 multiple times (more than 500) and results in a high cache hit rate (around 80-90%). Look at the assembly code to analyze what could be the reason for the same.

Type: **Semantic Analysis**

24. For the memory access with PC 0x4090c3 and address 0x3a4c39acd9d, the parrot replacement policy results in a cache hit. Looking at the assembly code, what might be the reason for this?

Type: **Semantic Analysis**

25. The miss rate for PC 0x4037ba on the mcf workload with PARROT replacement policy is 3.50%. Looking at the assembly code, what might be the reason for this?

Type: **Semantic Analysis**